Technical Approach Report

Architecture, implementation, evaluation, controls and roadmap

Prepared for Cynbit Technologies

Prepared by Surya Prakash Polkam • 20 October 2026

Assessment deliverable 03 | Confidential submission draft

Technical Approach Report

Architecture, implementation, evaluation, controls and roadmap

Prepared for Cynbit Technologies • Prepared by Surya Prakash Polkam • 20 October 2026

Assessment deliverable 03 | Confidential submission draft

1. Technical objective

LeadIQ explores assisted intake of project inquiries. It combines LLM-based structured extraction, schema validation, a small ML readiness classifier, deterministic guardrails, a web UI, a REST API and persistence. The intended decision remains with a human reviewer.

2. Architecture

Layer	Technology	Responsibility
UI	Streamlit	Inquiry input, result review, draft display, save action and recent records.
LLM	Gemini API via OpenAI-compatible client	Extract structured information from free text.
Validation	Pydantic	Validate required fields and structured result shape.
ML baseline	scikit-learn TF-IDF + Logistic Regression	Predict one of three readiness categories.
Guardrails	Python rules	Override some labels for unclear service or missing requirements/budget/timeline.
API	FastAPI + Uvicorn	Expose health, analysis and saved analysis endpoints.
Persistence	SQLAlchemy; SQLite default, MySQL configurable	Store analysis and readiness output.
Tests	pytest	Check blank input, structured fields and readiness guardrails.

3. Data flow

– User submits inquiry text through Streamlit or FastAPI.

– analyze_inquiry() validates input length and calls Gemini when credentials/model are configured.

– The structured response is parsed as JSON and validated into InquiryAnalysis; a fallback path is available if the LLM is not configured or fails.

– predict_readiness() builds text from summary, category, requirements, budget and timeline, then applies a TF-IDF + Logistic Regression classifier.

– Guardrail rules override selected predictions for unclear service, missing requirements, budget or timeline.

– The UI shows output for human review; save_analysis() persists the result.

– FastAPI exposes a programmatic path to analysis and database-backed records.

4. ML methodology

The baseline uses TfidfVectorizer with unigrams and bigrams, lowercase text, and LogisticRegression(max_iter=1000, random_state=42). It trains on 16 manually authored synthetic examples: 6 Ready for discovery, 5 Needs clarification and 5 Low information. The model is rebuilt at prediction time, which is acceptable for a tiny prototype but inefficient for a service.

The classifier output is an experimental category, not a calibrated probability. Rules override the classifier when the service category is Unclear, requirements are empty, or budget/timeline is missing. These rules improve interpretability but do not make the system validated for business decisions.

5. Evaluation results

Measure	Observed result	Interpretation
Synthetic dataset	16 examples	Too small and artificial for generalisation claims.
Exploratory 3-fold CV accuracy	0.50	Half of held-out predictions correct in this small experiment.
Macro F1	0.41	Weak average performance across the three labels.
Low information precision / recall	0.50 / 0.20	Misses most examples in this class.
Needs clarification precision / recall	0.33 / 0.20	Misses most examples in this class.
Ready for discovery precision / recall	0.55 / 1.00	Finds all ready examples but also over-labels some others as ready.
Functional tests	6 passed	Functional tests do not establish predictive quality.

The cross-validation results are unstable because each fold contains very few examples. The values should be presented as exploratory evidence only. No conversion probability, production accuracy, ROI, or causal impact is claimed.

6. API and persistence

FastAPI exposes GET /health, POST /analyze and GET /analyses. The local development server is run with `python -m uvicorn api:app --reload`; Swagger documentation is available at `/docs`. The user reported that `/analyze` responded successfully. The database uses SQLAlchemy with SQLite by default and MySQL configurable through DATABASE_URL.

- Add authentication and authorization before network exposure.
- Add request size limits, rate limiting, timeout/retry policy and safer error responses.
- Use migrations, backups, least-privilege database credentials and retention controls.
- Add integration tests for database errors, API validation and concurrent requests.
- Add idempotency and audit logs for production CRM write-back.

7. Security and responsible AI

- Store secrets only in .env or a secret manager; never commit credentials.
- If an API key is exposed, revoke it and replace it.
- Use synthetic data in demos; review provider terms and obtain organizational approval before processing customer data.
- Treat inquiry content as untrusted input, not as system instructions.
- Preserve source text and avoid inventing absent budget/timeline values.
- Require human approval for external messages and any consequential action.
- Define data minimisation, retention, deletion and access policies.

8. Limitations

- No representative, independently labeled real-world evaluation dataset.
- LLM extraction quality has not been benchmarked field-by-field.
- No evidence of production load, uptime, latency, cost at scale or security penetration testing.
- No CRM write-back or cloud deployment is evidenced.
- No model versioning, monitoring or drift detection in the current baseline.
- API smoke test was reported successful but visual API evidence was not captured.

9. Roadmap

Milestone	Work	Acceptance criteria
M1 — Reproducible PoC	Freeze source/dependencies; deterministic tests; separate ML/rule reason codes.	All tests pass from clean setup; clear demo limitations.
M2 — Dataset quality	Build a consented/anonymised, reviewer-labelled dataset and annotation guide.	Predefined holdout; per-class metrics and error analysis.
M3 — API hardening	Auth, rate limits, timeouts, monitoring, audit logs and secret management.	Security and failure-mode review passed.
M4 — Shadow CRM pilot	Human-reviewed suggestions without automatic outbound actions.	Measured baseline comparison and override tracking.
M5 — Controlled release	Model/data versioning, drift review and rollback.	Business owner approves explicit quality and safety thresholds.

Conclusion

LeadIQ demonstrates an end-to-end prototype integrating LLM extraction, ML classification, deterministic guardrails, Streamlit, FastAPI and database persistence. Its strongest current contribution is the integrated workflow and transparent evaluation—not predictive performance. Representative labeled data, security hardening and real workflow validation are prerequisites for production.

References

– Cynbit official website — https://www.cynbit.in/

– Cynbit services — https://www.cynbit.in/services

– Cynbit contact / inquiry page — https://www.cynbit.in/contact

– FastAPI documentation — https://fastapi.tiangolo.com/

– scikit-learn model evaluation — https://scikit-learn.org/stable/modules/model_evaluation.html

– Google Gemini API OpenAI compatibility — https://ai.google.dev/gemini-api/docs/openai